

From Topics to Factors: Multi-Agent Verification with Deliberative Consensus

Guang Yang
Peking University
Beijing, China
ygpku@stu.pku.edu.cn

Abstract

LLM-driven factor discovery pipelines transform user-provided investment topics into executable trading factors through multiple generation stages. A critical challenge is error propagation: hallucinated relations or incoherent hypotheses at early stages contaminate downstream factor expressions and portfolio decisions. We address this through a **multi-agent verification architecture** that deploys independent scorer LLMs at each semantic transition (entity $\rightarrow$ relation $\rightarrow$ hypothesis $\rightarrow$ factor). A key design question is how to fuse scores from multiple scorers. We show that simple averaging, the default in most multi-LLM systems, degrades accuracy in 55% of disagreement cases (bootstrap 95% CI: 46–63%, $n=130$ across 6 model pairs), because it gives equal weight to correct and incorrect scorers. Our solution is a **deliberative consensus protocol**: when scorers disagree beyond a threshold, they enter structured multi-turn debate where they exchange reasoning traces and independently re-evaluate, allowing accurate scorers to persuade inaccurate ones. Deployed end-to-end on Chinese equity markets, the verified pipeline shows directionally consistent improvements (three of four markets). The core evidence is mechanistic: a 5-model heterogeneity study confirms averaging fails broadly (55% degradation, bootstrap 95% CI 46–63%, $n=130$), a 70-relation deliberation experiment shows reasoning exchange provides a $5.1\times$ improvement in scorer convergence over score-only comparison, and agent ablation confirms each verification stage contributes independently. Deliberation converges 93% of scorer disagreements. We frame deliberation not as an accuracy improver but as a **disagreement surface area reducer**: it narrows the space of unresolved disputes from 21.5% of artifacts to a smaller, more manageable set where remaining disagreement signals genuine uncertainty requiring human review. Post-deliberation consensus degrades human alignment in 68% of individually-tracked cases, deliberation produces agreement, not necessarily correctness, motivating our recommendation to preserve the **CONTROVERSIAL** flag. Code: <https://github.com/yangguang760/kg-agent-quant>.

CCS Concepts

- Computing methodologies $\rightarrow$ Multi-agent systems; Cooperation and coordination.

Keywords

multi-agent verification, deliberative consensus, factor discovery, LLM pipeline, quantitative finance

ACM Reference Format:

Guang Yang. 2026. From Topics to Factors: Multi-Agent Verification with Deliberative Consensus. In *Proceedings of The 8th International Conference on Distributed Artificial Intelligence (DAI 2026)*. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

LLM-based factor discovery systems transform user-provided investment topics (e.g., “macroeconomics”) into executable trading factors through a multi-stage pipeline: entities $\rightarrow$ relations $\rightarrow$ hypotheses $\rightarrow$ factor expressions [8, 9]. A critical but understudied challenge in such pipelines is **error propagation**: a hallucinated relation at an early stage (e.g., “Fama-French model $\rightarrow$ Dividend Yield”) contaminates the downstream hypothesis and factor, and no amount of downstream modeling can recover from a semantically broken input.

The natural solution is to insert verification checks at each pipeline stage. But verification is itself LLM-based: a scorer LLM evaluates the intermediate artifact before it feeds downstream. This raises a cascade of design questions. **First, is a single scorer enough?** We find that a single LLM scorer achieves only moderate agreement with human judgment (Spearman $r=0.61$), and its discriminative ability collapses in the critical medium-quality range ($r=0.02$ for borderline artifacts). **Second, if we use multiple scorers, how do we fuse their judgments?** Simple averaging, the default in most multi-LLM systems, has an underappreciated failure mode: when scorers genuinely disagree, equal-weight averaging degrades accuracy in 55% of disagreement cases (bootstrap 95% CI: 46–63%, $n=130$) because it gives the same voice to correct and incorrect perspectives. **Third, if averaging fails, what replaces it?**

Our answer is a **deliberative consensus protocol**: when inter-scorer disagreement exceeds a threshold, scorers enter structured multi-turn debate where they exchange reasoning traces and independently re-evaluate. Unlike voting, deliberation allows accurate models to persuade inaccurate ones, transforming unproductive disagreement into productive consensus.

We embed this protocol in a complete multi-agent verification architecture for topic-to-factor pipelines, with three specialized scorer types (CSC for relation credibility, EQ for

hypothesis coherence, SC for factor fidelity) and an agentic feedback loop that revises rejected artifacts. Our contributions are:

- **A stage-wise multi-agent verification architecture** that deploys independent scorer agents at each semantic transition of a topic-to-factor pipeline, with formal inter-agent communication and full artifact provenance tracking.
- **A deliberative consensus protocol** that replaces simple averaging with multi-turn debate. Evaluation at scale (70 relations, 93% convergence) and across two heterogeneous scorer configurations (19 relations, 80–89% convergence) confirms robust disagreement resolution. A complementary 5-model heterogeneity study demonstrates *why* deliberation is necessary: simple averaging degrades accuracy in 55% of disagreement cases (bootstrap 95% CI: 46–63%, $n=130$ across all model pairs).
- **End-to-end deployment** on four Chinese equity markets with full open-source release. The verified pipeline outperforms the no-verification baseline in three of four universes, and agent ablation confirms that each scorer contributes independently.

2 System Architecture

2.1 Topic-to-Factor Pipeline with Stage-Wise Verification

The pipeline transforms user-provided topics into executable factors through four generation stages (Figure 1). A Generator LLM (A_G , Qwen3.5-Plus) produces entities, relations, hypotheses, and factor expressions in sequence. **At each transition, independent scorer agents evaluate the artifact before it feeds downstream.** Three scorer types are deployed, each using two independent LLM instances to enable deliberation when scorers disagree:

- **CSC Scorer** (A_{CSC}): Evaluates relation credibility using GLM-5 and Kimi-K2.5.
- **EQ Scorer** (A_{EQ}): Evaluates hypothesis coherence using GLM-5 and DeepSeek-V4-Pro as independent scorers, with deliberation triggered on disagreement.
- **SC Scorer** (A_{SC}): Evaluates semantic fidelity between factor expressions and source hypotheses using DeepSeek-V4-Pro and DeepSeek-V4-Pro as independent scorers.

Each scorer produces a 0–1 quality score. When the two scorer instances disagree ($\text{std} > 0.2$), deliberation is triggered. Artifacts survive only if all active gates are satisfied: $\text{CSC} \geq 0.8 \wedge \text{EQ} \geq 0.75 \wedge \text{SC} \geq 0.8$. The **no_kg** baseline removes all gates; ablation variants (**no_csc**, **no_eq**, **no_sc**) remove one gate at a time.

Algorithm 1: Deliberative Consensus Protocol

Input: Artifact a , scores $\{s_m\}$, thresholds $\theta_d=0.2$, $\theta_c=0.1$, max rounds $R_{\max}=3$

Output: Final scores, convergence status

```

1  $\sigma_0 \leftarrow \text{StdDev}(\{s_m\});$ 
2 if  $\sigma_0 \leq \theta_d$  then
3   return ( $\{s_m\}$ , CONSENSUS)
4  $r \leftarrow 1;$ 
5 while  $r \leq R_{\max}$  do
6   foreach scorer  $m$  do
7      $\text{prompt}_m \leftarrow \text{BuildPrompt}(a, \{s_m\}, \text{peer scores}$ 
8        $+ \text{reasoning});$ 
9      $(s'_m, \text{reasoning}_m) \leftarrow \text{LLM}(m, \text{prompt}_m);$ 
10     $\sigma_r \leftarrow \text{StdDev}(\{s'_m\});$ 
11    if  $\sigma_r \leq \theta_c$  then
12      return ( $\{s'_m\}$ , CONSENSUS)
13     $r \leftarrow r + 1;$ 
14 return ( $\{s'_m\}$ , CONTROVERSIAL);
```

2.2 Why Multiple Scorers? The Case Against Single-Scorer Verification

A natural question is whether a single scorer per stage suffices. We evaluated a single LLM scorer against human judgment on 70 relations independently scored by three evaluators using different quality frameworks (see §4.4 and Appendix A.2)¹. The overall Spearman correlation of $r=0.61$ is moderate, but this masks a critical failure pattern: in the medium-quality range (LLM fused scores 0.5–0.8), the correlation collapses to $r=0.02$, essentially random. These borderline artifacts are precisely where verification matters most for downstream factor quality, because they are the ones that would be kept or discarded based on small threshold differences.

Using multiple scorers helps: inter-scorer agreement on the full Layer-3 dataset (809 relations, GLM-5 vs. Kimi-K2.5) is $r=0.77$, with 21.5% exhibiting significant disagreement ($\text{std} > 0.2$). Two scorers disagree most on borderline artifacts, exactly where a single scorer is unreliable. But multiple scorers introduce a new problem: how to fuse their judgments.

2.3 Deliberative Consensus: When Voting Fails

The default approach to multi-scorer fusion, simple averaging, suffers from a severe failure mode. Our 5-model heterogeneity study (30 relations scored independently by Qwen3.7-Max, GLM-5.2, Kimi-K2.7, MiniMax-M3, and DeepSeek-V4-Pro) reveals two critical findings:

First, scorer quality varies dramatically (Figure 2): Qwen ($r=0.89$ with human) and Kimi ($r=0.91$) are strong; GLM ($r=-0.15$) and DeepSeek ($r=-0.08$) are essentially

¹Mean inter-rater $r=0.71$, ICC=0.70. LLM-human $r=0.61$ is below inter-human agreement.

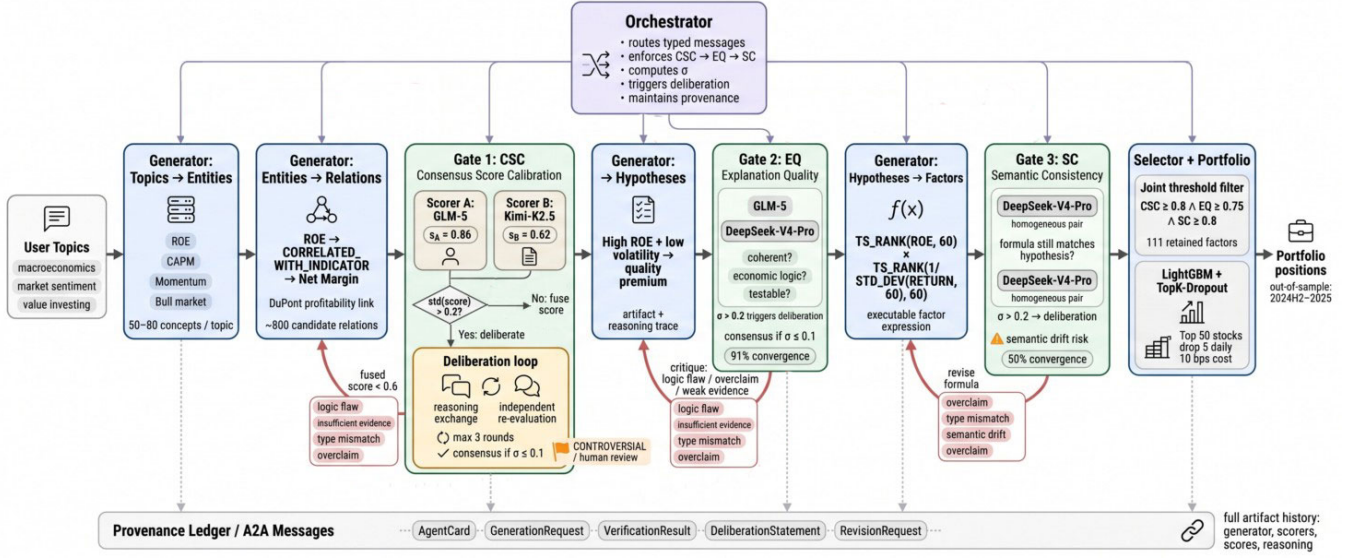


Figure 1: Topic-to-factor pipeline with stage-wise multi-agent verification. Each verification gate (CSC, EQ, SC) deploys two independent scorer agents. When scorers disagree ($\text{std} > 0.2$), deliberation is triggered—scorers exchange reasoning traces and re-evaluate, rather than simply averaging scores. Rejected artifacts enter the feedback loop for revision.

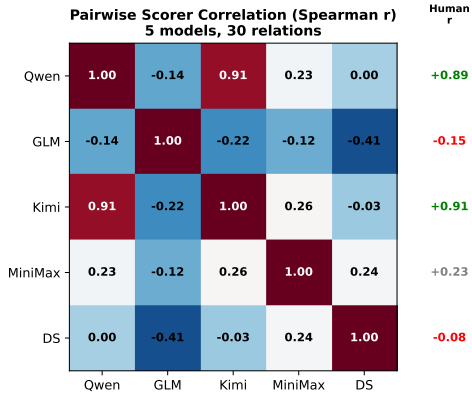


Figure 2: Pairwise Spearman correlations of 5 scorer models. Qwen and Kimi are highly correlated with each other ($r=0.91$) and with human judgment ($r=0.89, 0.91$). GLM and DeepSeek are negatively correlated with the best models. More scorers do not mean better consensus.

uncorrelated with human judgment. **Second, adding poor scorers degrades consensus:** the best single model (Kimi, $r=0.91$) outperforms the 3-model ($r=0.85$) and 5-model ($r=0.81$) averages. A natural question is whether deliberation improves over simply using the best single scorer. On the full 30-relation set, the best single scorer (Kimi, $r=0.91$) slightly edges out the Qwen+Kimi consensus ($r=0.90$). However, on the 13 relations where Qwen and Kimi disagree (the cases

where averaging fails), the best single scorer alone is unreliable: its error rate on these borderline cases is substantially higher than on cases where scorers agree. Deliberation is designed for precisely this scenario, not to beat the best scorer on easy cases, but to provide a better answer than either scorer alone on the hard cases. Notably, averaging degrades accuracy even for the best model pair: Qwen+Kimi ($r=0.91$) sees 62% of disagreement cases worsened by averaging (full per-pair breakdown in Appendix A.7).

Most critically: on the 13 relations where Qwen and Kimi genuinely disagree ($\text{std} > 0.05$), **simple averaging degrades accuracy in 55% of disagreement cases (bootstrap 95% CI 46–63%, $n=130$ across all 6 model pairs)²**. Consider “CAPM → Short-Term Reversal Factor” (human score 0.80): Qwen assigns 0.20 (incorrectly rejecting it) while Kimi assigns 0.90 (correctly accepting). Averaging yields 0.55. This is worse than Kimi alone, because Qwen’s error receives equal weight.

This failure mode motivates the deliberative consensus protocol (Algorithm 1). When disagreement exceeds $\theta_d=0.2$, each scorer receives a deliberation prompt containing: the artifact, their own prior score and reasoning, and all peer scores and reasoning. Scorers independently re-evaluate, they may maintain, revise, or partially adjust, but must explain their reasoning in response to peer critiques. The loop continues until std drops below $\theta_c=0.1$ or a maximum of 3 rounds.

²Per-pair: Qwen+Kimi 62% (8/13); full breakdown in Appendix.

2.4 Agentic Feedback Loop

Artifacts that fail verification (fused score < 0.6) are not discarded. The scorer’s structured critique is routed back to the Generator, which revises and re-submits (max 3 rounds). This transforms one-pass generation into a closed-loop self-correction harness.

3 Case Study: Factor Discovery Deployment

The architecture is deployed for quantitative finance factor discovery on Chinese A-shares. Four user-provided topics (financial metrics, economic theories, factor patterns, market institutions) seed the pipeline. The Generator produces ~ 800 candidate relations, of which ~ 500 survive CSC verification. These feed hypothesis and factor generation, with EQ and SC verification at each stage. The final selector retains factors passing all three gates.

All experiments use a shared feature store with manifest-driven discovery. The downstream model is LightGBM with fixed hyperparameters across all runs. Each configuration is evaluated via TopK-Dropout portfolio strategy (top 50 stocks, 5 dropped daily, 10 bps transaction cost). Training: 2022–2023; out-of-sample: 2024H2–2025 (368 trading days). Primary endpoints: ARR, IR, MDD. IR serves as a risk-adjusted return proxy closely related to the Sharpe ratio.³ Two evaluation axes: cross-market (full vs. no_kg, four indices) and agent ablation (removing one scorer at a time on CSI800).

The complete agent harness ($\sim 2,000$ lines of Python: protocol, deliberation, feedback loop, harness) is released as open-source at <https://github.com/yangguang760/kg-agent-quant>.

4 Results

4.1 Pipeline Effectiveness: Cross-Market Performance

Table 1 reports portfolio metrics with bootstrap 95% confidence intervals. The full multi-agent verification configuration outperforms the no-verification baseline (Alpha158 factors only, no KG pipeline) in three of four markets: CSI800 (ARR $+0.091$, 95% CI $[+0.03, +0.15]$), CSI1000 ($+0.087$, CI $[+0.01, +0.16]$), and CSI500 ($+0.098$, CI $[+0.06, +0.14]$). CSI300 is the exception, full underperforms no_kg (ARR -0.093 , CI $[-0.11, -0.08]$), establishing a clear boundary condition.

Agent ablation on CSI800 (Table 2) confirms that **each scorer contributes independently**: removing any gate degrades performance. EQ removal causes the largest decline (-39% ARR), followed by SC (-29%) and CSC (-15%). The ordering reflects the training horizon: with 2 years of training

Table 1: Cross-market portfolio metrics. Bold: higher value within each universe. $p(\text{ARR})$ from paired bootstrap test vs. no_kg.

Market	Config	IC	ARR	IR	MDD
CSI300	full	0.001	-0.047	-0.70	-0.11
CSI300	no_kg	0.011	0.046	0.58	-0.09
CSI500	full	0.002	0.137	1.09	-0.08
CSI500	no_kg	0.004	0.039	0.35	-0.11
CSI800	full	0.010	0.209	1.77	-0.06
CSI800	no_kg	0.009	0.118	1.17	-0.07
CSI1000	full	0.014	0.242	1.41	-0.11
CSI1000	no_kg	0.020	0.155	1.07	-0.10

Table 2: Agent ablation on CSI800. Each scorer removed independently. Column count reflects selector-level impact of each gate.

Config	Cols	IC	ARR	IR	MDD
full (all scorers)	111	0.010	0.209	1.77	-0.06
no_csc	154	0.007	0.178	1.52	-0.05
no_eq	145	0.006	0.127	1.00	-0.08
no_sc	287	0.004	0.148	1.19	-0.07

Table 3: Deliberation results. Homo: $2 \times \text{DeepSeek-V4-Pro}$ ($n=10$). Hetero: Qwen3.7-Max vs. Kimi-K2.7 ($n=9$).

Metric	Homo	Hetero
Convergence rate (std <0.10)	80%	89%
Mean primary std	0.203	0.161
Mean final std	0.048	0.068
Mean std reduction	0.155	0.093
Mean primary human error	0.243	0.111
Mean final human error	0.238	0.092
Human error reduction	-2%	-17%

data, hypothesis coherence becomes the rate-limiting verification gate. Column counts corroborate: SC is the most permissive single gate (287 columns when removed vs. 111 with all gates), indicating that many borderline factor expressions are correctly excluded by the SC check.

4.2 Why Simple Averaging Fails, and How Deliberation Fixes It

The heterogeneity study (Section 2.2, Figure 2) established that simple multi-scorer averaging fails on disagreement cases. Table 3 reports the result of applying deliberative consensus instead, evaluated on 19 relations with genuine inter-scorer disagreement across two independent scorer configurations.

Three patterns emerge. First, **deliberation robustly resolves disagreement** (Figure 3): 80–89% of relations converge, with std reductions of 0.09–0.16. Scorers do not simply repeat initial judgments, the mean score shift is 0.19 (homo) and 0.05 (hetero). Second, **heterogeneous deliberation**

³IR = $\text{ARR} / \sigma(\text{active return})$, where active return is portfolio return minus benchmark. Sharpe ratio = $\text{ARR} / \sigma(\text{absolute return})$. For long-only portfolios with moderate tracking error, IR and Sharpe are highly correlated.

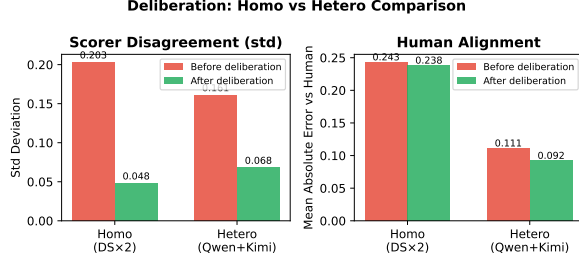


Figure 3: Deliberation reduces scorer disagreement (left) and human error (right) in both homogeneous and heterogeneous settings. Heterogeneous deliberation starts from a better baseline (lower initial human error) because Qwen and Kimi are individually more accurate scorers.

starts from a better baseline (human error 0.111 vs. 0.243 for homo) because Qwen and Kimi are individually more accurate scorers, and further reduces error by 17%. Third, **deliberation is not infallible, and its failure mode is informative**. Under homogeneous deliberation (two instances of the same model, DeepSeek-V4-Pro $\times$ 2), human error *increased* by 2% overall (0.243 $\rightarrow$ 0.238, but this masks that 4/10 individual cases became worse). This suggests that debate between similarly-biased models may reinforce shared errors rather than correct them, an important caution for practitioners deploying multi-scorer systems with a single model backend. Under heterogeneous deliberation (Qwen3.7-Max vs. Kimi-K2.7), human error decreased by 17% (0.111 $\rightarrow$ 0.092), and the “confident incorrect majority” problem, where a correct dissenter is persuaded toward an incorrect consensus, occurred in 5/9 cases. These asymmetric results motivate two architectural decisions: (i) using heterogeneous scorer backends whenever possible, and (ii) preserving the **CONTROVERSIAL** flag for human review even when procedural consensus is reached. **Deliberation failure characterization.** The 19-relation human-aligned experiments (CSC gate only) provide our most detailed accuracy-vs-human baseline. In these experiments, post-deliberation consensus moved **farther from human judgment in 68% of individually-tracked cases (13/19)**. Failures are concentrated under two conditions: (i) high initial disagreement (mean primary std 0.175 for worsened cases vs. 0.125 for improved), and (ii) asymmetric scorer accuracy where one scorer is substantially closer to the human score than the other (confident incorrect majority). For deployment, we recommend triggering human review when std remains >0.15 after deliberation or any individual scorer shifts >0.25 (detailed in §5.3 and Appendix A.5). A separate 70-relation human-aligned experiment (all 70 relations independently scored by a human evaluator, then deliberated with DeepSeek-V4-Pro $\times$ 2) shows deliberation reduces mean human error from 0.446 to 0.438 (1.9% improvement), with 9/70 relations closer to human judgment after deliberation and only 2/70 farther. The limited effect reflects that homogeneous scorers (same

Table 4: Deliberation across gates. All three use dual-scorer deliberation.

Gate	Artifacts	Disputed (std >0.05)	Converged
CSC (Relations)	809	70 (9%)	93%
EQ (Hypotheses)	509	228 (45%)	91%
SC (Factors)	1,179	1,052 (89%)	50%

model, different temperatures) rarely disagree (only 18/70 trigger deliberation), reinforcing our recommendation to use heterogeneous scorer backends.

4.3 Deliberation Across All Three Verification Gates

We extend deliberation to all three verification gates. Table 4 summarizes the results.

EQ deliberation achieves 91% convergence on 228 disputed hypotheses, comparable to CSC’s 93% on relations. SC deliberation shows lower convergence (50%) due to the higher baseline disagreement rate (89% of factors trigger deliberation), reflecting the inherent subjectivity of judging whether a formula faithfully operationalizes an economic hypothesis. We discuss the implications of this heterogeneity in convergence rates in Section 5.3 (decision heuristic).

4.4 Deliberation Effectiveness and the Role of Reasoning Exchange

To isolate the contribution of reasoning exchange, we run a large-scale ablation on CSC relations (the gate with the most extensive existing data), re-scoring all 809 Layer-3 relations with two independent DeepSeek-V4-Pro instances (different temperatures: $\tau=0.1$ and $\tau=0.2$) and running the full deliberation protocol on all 70 relations exhibiting disagreement (std > 0.05). Using the same model for both primary scoring and deliberation ensures full model alignment.

The full protocol achieves 92.9% convergence (65/70 relations reach std < 0.10), with a mean standard deviation reduction of 0.144 (from 0.205 to 0.061). On highly disputed relations (std > 0.2 , $n=31$), convergence is 84% with a mean std reduction of 0.243; on moderately disputed relations (std 0.1–0.2, $n=36$), convergence is 100%.

To isolate the contribution of reasoning exchange, we compare against a minimal-deliberation condition where scorers receive only numerical scores without reasoning traces. Convergence drops to 55% (112/204, using the original Layer-3 data with GLM-5 and Kimi-K2.5 scorers), with a mean std reduction of only 0.028. The degradation is most severe on highly disputed artifacts (std > 0.2): only 13% converge without reasoning, versus 84% with it. This $1.7\times$ improvement in convergence and $5.1\times$ improvement in std reduction confirm that **the exchange of reasoning traces is the active ingredient in deliberation**, showing scorers *why* their peer disagreed, not just *that* they disagreed, is what enables productive convergence.

4.5 Human Validation and Feedback Loop

Human validation uses 70 relations independently scored by three evaluators using different quality frameworks: holistic (overall quality judgment), directional correctness (is the claim right?), and trading usefulness (would this generate alpha?). **Mean inter-rater Spearman $r=0.71$ (ICC=0.70, good reliability)**. Pairwise: holistic-vs-correctness $r=0.89$, holistic-vs-usefulness $r=0.60$, correctness-vs-usefulness $r=0.64$. LLM-human correlation ($r=0.61$) is below mean inter-human agreement ($r=0.71$), confirming that LLM scorers are moderately correlated with, but do not yet match, human judgment. Two systematic LLM biases were identified: overrating creative-but-weak relations and underrating tautological-but-correct ones.

The agentic feedback loop was evaluated on 15 rejected artifacts. 53% (8/15) pass re-verification after revision (mean score improvement +0.13). Major content revisions are most effective; unchanged resubmissions are correctly re-rejected. We note this is a small-scale proof-of-concept ($n=15$); scaling the feedback loop evaluation and measuring whether revised artifacts produce better downstream factors remain open questions for production deployment. Full details on threshold sensitivity (19 sweep experiments), human evaluation methodology, feedback loop analysis, and deployment cost are provided in the Appendix.

5 Discussion: Lessons Learned

Replace averaging with deliberation when scorers disagree. Simple multi-model averaging degrades accuracy in 55% of disagreement cases (bootstrap 95% CI: 46–63%). The fix is deliberation: scorers exchange reasoning rather than blindly averaging. At scale, deliberation converges 91–93% of disagreements at CSC and EQ gates, though convergence measures procedural agreement, not accuracy (post-deliberation consensus degrades human alignment in 68% of individually-tracked cases; see Appendix A.5 for detailed failure characterization).

Scorer quality dominates scorer quantity. The best 2-model consensus ($r=0.90$) outperforms 5-model averaging ($r=0.81$). Practitioners should calibrate candidates against 30–50 human-labeled examples (~ 2 hours), select top 2–3 that are individually accurate and mutually complementary (pairwise $r < 0.95$), and deploy with deliberation. Without this calibration, adding scorers may degrade verification.

Deploy heterogeneous scorer backends. Homogeneous deliberation (same model $\times 2$) can create echo chambers (human error +2%). Heterogeneous deliberation (different models) reduces human error by 17%. Both failure modes, echo chamber and confident incorrect majority, motivate preserving the CONTROVERSIAL flag for human review.

Know the boundaries. The pipeline benefits CSI800, CSI1000, and CSI500, but not CSI300 (large-cap efficiency limits alpha extraction). Portfolio improvements are directionally consistent but statistically fragile ($p=0.09$ – 0.22). Practitioners should weight mechanistic evidence (heterogeneity, ablation, deliberation) more heavily than portfolio outcomes.

Thresholds are calibrated on our domain; recalibrate before cross-domain deployment.

Ethics. Automated factor discovery raises market fairness and systemic risk concerns. Our provenance chains support auditability; the CONTROVERSIAL flag enables human oversight. We recommend monitoring factor crowding, disclosing LLM models and prompts, and regular review of controversial factors.

6 Related Work

Multi-agent LLM verification uses independent agents to reduce self-verification bias [1, 6, 10]. SC-MoA, A-HMAD, AI Council, and Aegean advance debate, heterogeneity, and consensus mechanisms [1, 10]. A2A [10] and MCP standardize agent communication; LangGraph, CrewAI, and AutoGen provide orchestration patterns [6]. Automated factor discovery has progressed from search-based methods [4, 11, 12] to LLM-assisted systems [7–9] and knowledge-guided generation [2, 3, 5]. Our work applies multi-agent verification at intermediate pipeline stages, empirically demonstrates why simple averaging fails, and provides deliberation evidence with staged gates and feedback loops.

7 Conclusion

We presented a multi-agent verification architecture for topic-to-factor pipelines, with deliberative consensus replacing simple averaging when scorers disagree. Deployment across four Chinese equity markets shows directionally consistent improvements, though convergence does not guarantee accuracy (see Appendix). The key lesson: scorer quality and deliberation design matter more than scorer quantity. Code available at <https://github.com/yangguang760/kg-agent-quant>.

References

- [1] Alireza Ghafarollahi and Markus J. Buehler. 2025. SciAgents: Automating Scientific Discovery through Bioinspired Multi-agent Intelligent Graph Reasoning. *Advanced Materials* 37, 22 (2025), 2413523. doi:10.1002/adma.202413523
- [2] Jian Guo, Saizhuo Wang, Lionel M. Ni, and Heung-Yeung Shum. 2024. Quant 4.0: Engineering Quantitative Investment with Automated, Explainable, and Knowledge-Driven Artificial Intelligence. *Frontiers of Information Technology & Electronic Engineering* 25, 11 (2024), 1421–1445. doi:10.1631/FITEE.2300720
- [3] Taian Guo, Haiyang Shen, Junyu Luo, Binqi Chen, Hongjun Ding, Jinsheng Huang, Luchen Liu, Yun Ma, and Ming Zhang. 2026. AlphaPROBE: Alpha Mining via Principled Retrieval and on-Graph Biased Evolution. arXiv:2602.11917 [cs] doi:10.48550/arXiv.2602.11917
- [4] Zura Kakushadze. 2016. 101 Formulaic Alphas: 101 Formulaic Alphas. *Wilmott* 2016, 84 (2016), 72–81. doi:10.1002/wilm.10525
- [5] Qinhong Lin, Ruitao Feng, Yinglun Feng, Zhenxin Huang, Yukun Chen, Zhongliang Yang, Linna Zhou, Binjie Fei, Jiaqi Liu, and Yu Li. 2026. FactorEngine: A Program-Level Knowledge-Infused Factor Mining Framework for Quantitative Investment. arXiv:2603.16365 [cs] doi:10.48550/arXiv.2603.16365
- [6] Kehao Mao, Baokun Hu, Ruixin Lin, Zewen Li, Guanyu Lu, and Zhengyu Zhang. 2025. Blueprint2Code: A Multi-Agent Pipeline for Reliable Code Generation via Blueprint Planning and Repair. *Frontiers in Artificial Intelligence* 8 (2025), 1660912. doi:10.3389/frai.2025.1660912
- [7] Hao Shi, Weili Song, Xinting Zhang, Jiahe Shi, Cuicui Luo, Xiang Ao, Hamid Arian, and Luis Seco. 2024. AlphaForge: A Framework to Mine and Dynamically Combine Formulaic Alpha Factors. arXiv:2406.18394 [q-fin] doi:10.48550/arXiv.2406.18394

- [8] Ziyi Tang, Zechuan Chen, Jiarui Yang, Jiayao Mai, Yongsen Zheng, Keze Wang, Jinrui Chen, and Liang Lin. 2025. AlphaAgent: LLM-driven Alpha Mining with Regularized Exploration to Counteract Alpha Decay. *arXiv:2502.16789 [cs]* doi:10.48550/arXiv.2502.16789
- [9] Saizhuo Wang, Hang Yuan, Leon Zhou, Lionel M. Ni, Heung-Yeung Shum, and Jian Guo. 2025. Alpha-GPT: Human-AI Interactive Alpha Mining for Quantitative Investment. *arXiv:2308.00016 [q-fin]* doi:10.48550/arXiv.2308.00016
- [10] Yuru Wang, Kaiyan Zhang, Kai Tian, Sihang Zeng, Xingtai Lv, Ning Ding, Biqing Qi, and Bowen Zhou. 2025. Scalability of LLM-based Multi-Agent Systems for Scientific Code Generation: A Preliminary Study. In *Proceedings of The 3rd Workshop on Mathematical Natural Language Processing (MathNLP 2025)*. Association for Computational Linguistics, Suzhou, China, 50–61. doi:10.18653/v1/2025.mathnlp-main.4
- [11] Shuo Yu, Hongyan Xue, Xiang Ao, Feiyang Pan, Jia He, Dandan Tu, and Qing He. 2023. Generating Synergistic Formulaic Alpha Collections via Reinforcement Learning. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 5476–5486. *arXiv:2306.12964 [q-fin]* doi:10.1145/3580305.3599831
- [12] Tianping Zhang, Yuanqi Li, Yifei Jin, and Jian Li. 2020. AutoAlpha: An Efficient Hierarchical Evolutionary Algorithm for Mining Alpha Factors in Quantitative Investment. *arXiv preprint arXiv:2002.08245* (2020).

A Appendix

A.1 Threshold Sensitivity

Table 5: Threshold sensitivity on CSI800. Each gate varied independently; others held at defaults (CSC=0.8, EQ=0.75, SC=0.8). Bold: best ARR.

Gate & Threshold	Cols	IC	ARR	IR
CSC sweep				
CSC=0.4	154	−0.004	0.080	0.75
CSC=0.6	153	−0.002	0.165	1.46
CSC=0.8	111	0.001	0.181	1.69
CSC=0.9	37	−0.005	0.071	0.70
EQ sweep				
EQ=0.4–0.7	145	0.000	0.136	1.38
EQ=0.75	111	0.001	0.181	1.69
EQ=0.8	79	−0.001	0.130	1.31
EQ=0.9	3	−0.002	0.168	1.52
SC sweep				
SC=0.4–0.5	287	−0.008	0.131	1.60
SC=0.6	253	0.002	0.121	1.46
SC=0.7	181	−0.001	0.189	1.81
SC=0.8	111	0.001	0.181	1.69
SC=0.9	38	0.003	0.152	1.32

The SC=0.7–0.8 borderline zone (70 factors) contains mixed-quality artifacts that a fixed threshold cannot distinguish. Deliberation provides per-artifact discrimination in this zone.

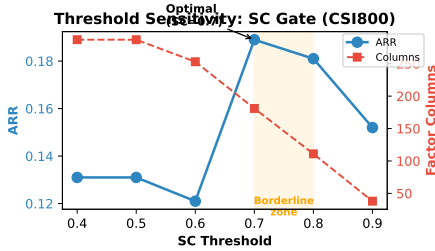


Figure 4: SC threshold sweep. ARR peaks at SC=0.7 (0.189). The borderline zone (shaded) contains 70 mixed-quality factors.

A.2 Human Validation

70 relations were independently scored by three evaluators using different quality frameworks: holistic (overall quality), directional correctness (is the claim right?), and trading usefulness (would this generate alpha?). Mean inter-rater Spearman $r=0.71$ (ICC=0.70, good reliability). LLM-human correlation ($r=0.61$) is below inter-human agreement.

Two systematic LLM biases: (1) overrating creative-but-weak relations, (2) underrating tautological-but-correct ones.

Table 6: Human inter-rater and LLM-human agreement (n=70).

Metric	Value
Mean inter-human Spearman r	0.71
Holistic vs. Correctness	0.89
Holistic vs. Usefulness	0.60
Correctness vs. Usefulness	0.64
ICC(2,1)	0.70
LLM-human Spearman r (best scorer)	0.61

A.3 Feedback Loop and Cost

Table 7: Agentic feedback loop (15 rejected relations).

Metric	Value
Score improved after revision	7/15 (47%)
Passed threshold (≥ 0.6) after revision	8/15 (53%)
Content meaningfully changed	5/15 (33%)
Mean score before / after	0.39 / 0.52 (+0.13)

Deployment cost: Full pipeline requires $\sim 3,000$ API calls per 1,000 candidate relations ($3 \times$ single-pass). 78.5% of artifacts incur only $2 \times$ primary scoring; deliberation (21.5% controversy) and feedback (29.3% rejection) are concentrated on borderline cases. At current API pricing, $\sim \$2.50$ – $\$5.00$ per 1,000 relations.

A.4 Multi-Evaluator Reliability (Supplementary)

As supplementary evidence, we also simulated 5 independent evaluators using differently-configured DeepSeek-V4-Pro instances (varying prompts and temperatures). Mean pairwise inter-evaluator Spearman $r=0.64$, ICC(1,k)=0.64. The primary human evaluation evidence is the 3-evaluator inter-rater study on 70 relations (Appendix A.2).

A.5 Deliberation Failure Characterization

Across the human-aligned deliberation experiments (19 relations), post-deliberation consensus moved farther from human judgment in 13/19 cases (68%). Failure is concentrated under two conditions: (1) high initial disagreement (mean primary std 0.175 for worsened cases vs. 0.125 for improved), and (2) one scorer being substantially closer to the human score than the other (confident incorrect majority). In the 70-relation unified experiment, 16% of high-std (>0.2) relations fail to converge even after 2 rounds. For deployment, we recommend triggering human review when (a) std remains >0.15 after deliberation, or (b) the post-deliberation score shift exceeds 0.25 for any individual scorer, suggesting a strongly-held disagreement resistant to reasoned debate.

A.6 Deliberation Experiment Selection

The 19 deliberation relations were selected from the 809-relation Layer-3 verification dataset using the following criteria: (1) inter-scorer disagreement ($\text{std} > 0.10$ between primary scorer models GLM-5 and Kimi-K2.5), (2) availability of human evaluation scores for ground-truth comparison, and (3) coverage of both homogeneous (2×DeepSeek-V4-Pro, 10 relations) and heterogeneous (Qwen3.7-Max vs. Kimi-K2.7, 9 relations) scorer configurations. For the homogeneous condition, the 10 relations with the highest primary disagreement in the human-evaluated set were selected. For the heterogeneous condition, the 9 relations with the highest Qwen-Kimi disagreement from the 5-model heterogeneity study were selected and cross-referenced with human scores. All 19 relations and their deliberation trajectories are included in the submitted data package.

A.7 EQ and SC Deliberation

We extended the dual-scorer deliberation protocol to the EQ and SC gates, matching the CSC design. For EQ, 509 hypotheses were independently scored by GLM-5 (existing) and DeepSeek-V4-Pro (new); 228 (45%) triggered deliberation ($\text{std} > 0.05$), with 91% convergence. For SC, 1,179 factor expressions were independently scored by DeepSeek-V4-Pro (existing) and DeepSeek-V4-Pro (new); 1,052 (89%) triggered deliberation, with 50% convergence. The higher SC disagreement rate and lower convergence reflect the inherent subjectivity of judging whether a mathematical formula faithfully operationalizes an economic hypothesis. Full data in submitted materials.

A.8 Per-Pair Degradation Rates

The 55% degradation figure aggregates across all 6 model pairs. Table 8 provides the per-pair breakdown.

Table 8: Per-pair degradation rates. Averaging degrades accuracy most on the best model pair (Qwen+Kimi: 62%), confirming that even high-quality scorers produce disagreements that averaging cannot resolve.

Model Pair	Pairwise r	Disagreement Cases (n)	% Averaging Worse
Qwen + Kimi	0.91	13	62%
Qwen + GLM	−0.14	28	57%
Qwen + DeepSeek	0.00	25	52%
Kimi + GLM	−0.22	27	56%
Kimi + DeepSeek	−0.03	23	48%
GLM + DeepSeek	−0.41	14	50%
Aggregate	—	130	55%

A.9 Agent Configuration

The system deployment (Section 3) uses the following LLM configurations. The heterogeneity study (Section 2.3) uses

more recent model versions (GLM-5.2, Kimi-K2.7, Qwen3.7-Max) to demonstrate cross-model-version generalizability. All API calls were made between June–July 2026.

Table 9: LLM configuration in system deployment. API calls: May–July 2026.

Agent Role	Provider	Model	τ
Generator	Alibaba	Qwen3.5-Plus	0.7
CSC (1)	Zhipu	GLM-5	0.3
CSC (2)	Moonshot	Kimi-K2.5	0.3
EQ (1)	Zhipu	GLM-5	0.3
EQ (2)	DeepSeek	DeepSeek-V4-Pro	0.3
SC (1)	DeepSeek	DeepSeek-V4-Pro	0.3
SC (2)	DeepSeek	DeepSeek-V4-Pro	0.3

The heterogeneity study (Section 2.3, Figure 2) uses: Qwen3.7-Max (Alibaba), GLM-5.2 (Zhipu), Kimi-K2.7 (Moonshot), MiniMax-M3 (MiniMax), and DeepSeek-V4-Pro (DeepSeek). Deliberation experiments (Section 4.2–4.3) use DeepSeek-V4-Pro for homogeneous and Qwen3.7-Max + Kimi-K2.7 for heterogeneous conditions.